Exponentials, Radicals, Polynomials & Rational Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the botanical conservatory is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 16. Reading the labeled points on the exponential curve, $(0, 2)$ gives the initial value $a = 2$, and $(1, 4)$ gives $2 \cdot b^1 = 4 \Rightarrow b = 2$. Therefore, $E(x) = 2 \cdot 2^x$, and $E(3) = 2 \cdot 2^3 = 16$.

2 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the regional bakery is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 135. Reading the labeled points on the exponential curve, $(0, 5)$ gives the initial value $a = 5$, and $(1, 15)$ gives $5 \cdot b^1 = 15 \Rightarrow b = 3$. Therefore, $E(x) = 5 \cdot 3^x$, and $E(3) = 5 \cdot 3^3 = 135$.

3 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the science museum is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 24. Reading the labeled points on the exponential curve, $(0, 3)$ gives the initial value $a = 3$, and $(1, 6)$ gives $3 \cdot b^1 = 6 \Rightarrow b = 2$. Therefore, $E(x) = 3 \cdot 2^x$, and $E(3) = 3 \cdot 2^3 = 24$.

4 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the ceramic studio is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 162. Reading the labeled points on the exponential curve, $(0, 6)$ gives the initial value $a = 6$, and $(1, 18)$ gives $6 \cdot b^1 = 18 \Rightarrow b = 3$. Therefore, $E(x) = 6 \cdot 3^x$, and $E(3) = 6 \cdot 3^3 = 162$.

5 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the public transit depot is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 32. Reading the labeled points on the exponential curve, $(0, 4)$ gives the initial value $a = 4$, and $(1, 8)$ gives $4 \cdot b^1 = 8 \Rightarrow b = 2$. Therefore, $E(x) = 4 \cdot 2^x$, and $E(3) = 4 \cdot 2^3 = 32$.

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6 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the art gallery is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 54. Reading the labeled points on the exponential curve, $(0, 2)$ gives the initial value $a = 2$, and $(1, 6)$ gives $2 \cdot b^1 = 6 \Rightarrow b = 3$. Therefore, $E(x) = 2 \cdot 3^x$, and $E(3) = 2 \cdot 3^3 = 54$.

7 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the university lecture hall is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

Answer: 40. Reading the labeled points on the exponential curve, $(0, 5)$ gives the initial value $a = 5$, and $(1, 10)$ gives $5 \cdot b^1 = 10 \Rightarrow b = 2$. Therefore, $E(x) = 5 \cdot 2^x$, and $E(3) = 5 \cdot 2^3 = 40$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $3^x = 81$, what is the value of x?

Answer: 4. $81 = 3^4$, so $3^x = 3^4 \Rightarrow x = 4$.

9 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the sports complex, in a calculation for the geology lab (case 40), when $(x + 4)(x + 1)$ is expanded, what is the coefficient of x?

Answer: 5. $(x + 4)(x + 1) = x^2 + 5x + 4$, so the coefficient of x is 5.

10 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the astronomical society, simplify $\frac{x^2 + 11x + 24}{(x + 3)}$, then evaluate the result at $x = 1$.

Answer: 9. $\frac{(x + 8)(x + 3)}{(x + 3)} = (x + 8)$ (for $x \neq -3$). At $x = 1$: 9.

11 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the film archive, in a calculation for the community theater (case 52), when $(x - 1)(x + 7)$ is expanded, what is the coefficient of x?

Answer: 6. $(x - 1)(x + 7) = x^2 + 6x - 7$, so the coefficient of x is 6.

12 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the mountain observatory, simplify $\frac{x^2 - 6x - 16}{(x + 2)}$, then evaluate the result at $x = -6$.

Answer: -14. $\frac{(x - 8)(x + 2)}{(x + 2)} = (x - 8)$ (for $x \neq -2$). At $x = -6$: -14.

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13 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the recycling center, in a calculation for the university lecture hall (case 60), solve for x: $\sqrt{x+8} = 4$.

Answer: 8. Square both sides: $x + 8 = 4^2 = 16 \Rightarrow x = 8$.

14 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), add: $(8x^2 - 4x + 3) + (2x^2 - 7x - 8)$

Answer: 2. Combine like terms to get $10x^2 - 11x - 5$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x > 0$, the expression $x \cdot \sqrt[2]{x^3}$ at the botanical conservatory can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

Answer: 5. Rewriting the radical using fractional exponents gives $x^1 \cdot x^{\frac{3}{2}} = x^{1+\frac{3}{2}} = x^{\frac{5}{2}}$. Thus $a = 5$.

16 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the municipal library, for $x \neq -3$, the rational expression $\frac{3x+15}{x+3}$ is equivalent to $3 + \frac{k}{x+3}$. What is the value of the constant k?

Answer: 6. Rewrite the numerator as $3(x+3) + 6$. Dividing by $x+3$ gives $3 + \frac{6}{x+3}$, so $k = 6$.

17 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A culture at the community theater begins with 300 cells and doubles every 5 hours according to $P(t) = 300(2)^{\frac{t}{5}}$. How many cells are in the culture after 20 hours?

Answer: 4800. Substitute $t = 20$ into the exponential growth model: $P(20) = 300(2)^{\frac{20}{5}} = 300(2)^4 = 4800$.

18 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x > 0$, the expression $x \cdot \sqrt[2]{x^6}$ at the regional bakery can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

Answer: 8. Rewriting the radical using fractional exponents gives $x^1 \cdot x^{\frac{6}{2}} = x^{1+\frac{6}{2}} = x^{\frac{8}{2}}$. Thus $a = 8$.

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19 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the coastal aquarium, for $x \neq -6$, the rational expression $\frac{6x+45}{x+6}$ is equivalent to $6 + \frac{k}{x+6}$. What is the value of the constant k?

Answer: 9. Rewrite the numerator as $6(x+6)+9$. Dividing by $x+6$ gives $6 + \frac{9}{x+6}$, so $k = 9$.

20 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A culture at the mountain observatory begins with 450 cells and doubles every 4 hours according to $P(t) = 450(2)^{\frac{t}{4}}$. How many cells are in the culture after 16 hours?

Answer: 7200. Substitute $t = 16$ into the exponential growth model:
$P(16) = 450(2)^{\frac{16}{4}} = 450(2)^4 = 7200$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the municipal library is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 27

b. 21

c. 84

d. 81 ✓

Reading the labeled points on the exponential curve, $(0, 3)$ gives the initial value $a = 3$, and $(1, 9)$ gives $3 \cdot b^1 = 9 \Rightarrow b = 3$. Therefore, $E(x) = 3 \cdot 3^x$, and $E(3) = 3 \cdot 3^3 = 81$.

22 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the community theater is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 36

b. 35

c. 32 ✓

d. 16

Reading the labeled points on the exponential curve, $(0, 4)$ gives the initial value $a = 4$, and $(1, 8)$ gives $4 \cdot b^1 = 8 \Rightarrow b = 2$. Therefore, $E(x) = 4 \cdot 2^x$, and $E(3) = 4 \cdot 2^3 = 32$.

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23 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the coastal aquarium is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 48 ✓

b. 24

c. 54

d. 51

Reading the labeled points on the exponential curve, $(0, 6)$ gives the initial value $a = 6$, and $(1, 12)$ gives $6 \cdot b^1 = 12 \Rightarrow b = 2$. Therefore, $E(x) = 6 \cdot 2^x$, and $E(3) = 6 \cdot 2^3 = 48$.

24 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the mountain observatory is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 18

b. 14

c. 56

d. 54 ✓

Reading the labeled points on the exponential curve, $(0, 2)$ gives the initial value $a = 2$, and $(1, 6)$ gives $2 \cdot b^1 = 6 \Rightarrow b = 3$. Therefore, $E(x) = 2 \cdot 3^x$, and $E(3) = 2 \cdot 3^3 = 54$.

25 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the urban bike-share station is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 112

b. 108 ✓

c. 36

d. 28

Reading the labeled points on the exponential curve, $(0, 4)$ gives the initial value $a = 4$, and $(1, 12)$ gives $4 \cdot b^1 = 12 \Rightarrow b = 3$. Therefore, $E(x) = 4 \cdot 3^x$, and $E(3) = 4 \cdot 3^3 = 108$.

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26 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the solar microgrid facility is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 40 ✓

b. 20

c. 45

d. 43

Reading the labeled points on the exponential curve, $(0, 5)$ gives the initial value $a = 5$, and $(1, 10)$ gives $5 \cdot b^1 = 10 \Rightarrow b = 2$. Therefore, $E(x) = 5 \cdot 2^x$, and $E(3) = 5 \cdot 2^3 = 40$.

27 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the agricultural cooperative is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 18

b. 19

c. 16 ✓

d. 8

Reading the labeled points on the exponential curve, $(0, 2)$ gives the initial value $a = 2$, and $(1, 4)$ gives $2 \cdot b^1 = 4 \Rightarrow b = 2$. Therefore, $E(x) = 2 \cdot 2^x$, and $E(3) = 2 \cdot 2^3 = 16$.

28 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

The graph of the exponential function $y = E(x) = a \cdot b^x$ modeling culture growth at the publishing house is shown in the xy-plane with two points labeled. Based on the graph, what is the value of $E(3)$?

a. 84

b. 81 ✓

c. 27

d. 21

Reading the labeled points on the exponential curve, $(0, 3)$ gives the initial value $a = 3$, and $(1, 9)$ gives $3 \cdot b^1 = 9 \Rightarrow b = 3$. Therefore, $E(x) = 3 \cdot 3^x$, and $E(3) = 3 \cdot 3^3 = 81$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the robotics club, in a calculation for the orchestra hall (case 12), simplify: $(b^8)^2$.

a. b^{16} ✓

b. b^{18}

c. b^{64}

d. b^{10}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(b^8)^2 = b^{8 \cdot 2} = b^{16}$

Therefore, the verified correct option is b^{16}.

30 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the culinary institute, in a calculation for the charter ferry service (case 13), subtract: $(2x^2 + 4x + 7) - (6x^2 + x + 3)$

a. $-4x^2 + 3x + 4$ ✓

b. $-2x^2 + 3x + 4$

c. $-4x^2 + 5x + 4$

d. $-4x^2 + 3x + 6$

Combine like terms to get $-4x^2 + 3x + 4$.

31 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), simplify: $(a^5)^3$.

a. a^{15} ✓

b. a^8

c. a^9

d. a^{17}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(a^5)^3 = a^{5 \cdot 3} = a^{15}$

Therefore, the verified correct option is a^{15}.

32 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the sports complex, in a calculation for the geology lab (case 15), subtract: $(6x^2 + 5x - 3) - (2x^2 + 2x - 5)$

a. $6x^2 + 3x + 2$

b. $4x^2 + 3x + 4$

c. $4x^2 + 5x + 2$

d. $4x^2 + 3x + 2$ ✓

Combine like terms to get $4x^2 + 3x + 2$.

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33 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the maple syrup farm, in a calculation for the honeybee apiary (case 16), add: $(8x^2 - 3x + 5) + (2x^2 - x - 3)$

a. $12x^2 - 4x + 2$

b. $10x^2 - 4x + 4$

c. $10x^2 - 2x + 2$

d. $10x^2 - 4x + 2$ ✓

Combine like terms to get $10x^2 - 4x + 2$.

34 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), simplify: $(y^8)^5$.

a. y^{14}

b. y^{42}

c. y^{13}

d. y^{40} ✓

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(y^8)^5 = y^{8 \cdot 5} = y^{40}$ Therefore, the verified correct option is y^{40}.

35 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), simplify: $(x^6)^9$.

a. x^{16}

b. x^{56}

c. x^{54} ✓

d. x^{15}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(x^6)^9 = x^{6 \cdot 9} = x^{54}$ Therefore, the verified correct option is x^{54}.

36 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the wind energy cooperative, in a calculation for the community orchard (case 19), simplify: $y^5 \cdot y^7$.

a. y^2

b. y^{12} ✓

c. $2y^{12}$

d. y^{35}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $y^5 \cdot y^7 = y^{5+7} = y^{12}$ Therefore, the verified correct option is y^{12}.

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37 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), simplify: $b^7 \cdot b^4$.

a. b^{28}

b. b^3

c. $2b^{11}$

d. b^{11} ✓

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $b^7 \cdot b^4 = b^{7+4} = b^{11}$ Therefore, the verified correct option is b^{11}.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the maple syrup farm, a radioactive isotope decays at a rate of 12% per day. If a sample initially has a mass of 80 grams, which function gives the mass $M(t)$ in grams remaining after t days?

a. $M(t) = 80(0.12)^t$

b. $M(t) = 80(0.88)^t$ ✓

c. $M(t) = 80(1.12)^t$

d. $M(t) = 80 - 0.12t$

Decaying at 12% means $100\% - 12\% = 88\% = 0.88$ remains each day: $M(t) = 80(0.88)^t$.

39 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the planetarium, simplify $\left(\frac{4x^4}{y^6}\right)^{-1/2}$ for positive x and y.

a. $\frac{y^3}{2x^2}$ ✓

b. $\frac{2x^2}{y^3}$

c. $\frac{y^3}{4x^2}$

d. $\frac{2y^3}{x^2}$

Negative exponent inverts the fraction: $\left(\frac{y^6}{4x^4}\right)^{1/2} = \frac{(y^6)^{1/2}}{4^{1/2}(x^4)^{1/2}} = \frac{y^3}{2x^2}$.

40 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 68), if $x^3 - 4x^2 + 5x - 2 = (x-1)(x-2)Q(x)$, what is $Q(x)$?

a. $x - 1$ ✓

b. $x + 1$

c. $x - 2$

d. $x + 2$

Notice $(x-1)(x-2) = x^2 - 3x + 2$. Divide $x^3 - 4x^2 + 5x - 2$ by $x^2 - 3x + 2$ gives quotient $x - 1$.

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41 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the wind energy cooperative, in a calculation for the community orchard (case 69), solve for x: $\sqrt{3x+1}+4=9$.

a. 7

b. 8 ✓

c. 9

d. 10

Subtract 4: $\sqrt{3x+1}=5$. Square both sides: $3x+1=25 \Rightarrow 3x=24 \Rightarrow x=8$.

42 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In a model for the marine research vessel, which of the following is equivalent to $\frac{2x}{x+3}+\frac{6}{x+3}$ for $x \neq -3$?

a. 2 ✓

b. $2x$

c. $\frac{2x+6}{(x+3)^2}$

d. $\frac{8x}{x+3}$

Combine numerators: $\frac{2x+6}{x+3}=\frac{2(x+3)}{x+3}=2$.

43 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

For $x>0$, the expression $x \cdot \sqrt[2]{x^5}$ at the planetarium can be rewritten in the form $x^{\frac{a}{2}}$. What is the value of the integer a?

a. 9

b. 7 ✓

c. 10

d. 5

Rewriting the radical using fractional exponents gives $x^1 \cdot x^{\frac{5}{2}}=x^{1+\frac{5}{2}}=x^{\frac{7}{2}}$. Thus $a=7$.

44 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

In an efficiency formula at the geology lab, for $x \neq -6$, the rational expression $\frac{2x+18}{x+6}$ is equivalent to $2+\frac{k}{x+6}$. What is the value of the constant k?

a. 6 ✓

b. 18

c. 12

d. 8

Rewrite the numerator as $2(x+6)+6$. Dividing by $x+6$ gives $2+\frac{6}{x+6}$, so $k=6$.

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45 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A culture at the regional airport shuttle begins with 450 cells and doubles every 4 hours according to $P(t) = 450(2)^{\frac{t}{4}}$. How many cells are in the culture after 16 hours?

a. 3600

b. 7650

c. 7203

d. 7200 ✓

Substitute $t = 16$ into the exponential growth model: $P(16) = 450(2)^{\frac{16}{4}} = 450(2)^4 = 7200$.

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